Discussion: Optimal Time-Consistent Macroprudential Policy

Bianchi & Mendoza (2018, *JPE*)

Guido Roveri

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Summary of the paper

Question: Can a collateral constraint generate financial crises, and what is the optimal macroprudential policy when the regulator cannot commit?

Model: small open economy; debt and working capital are capped at a fraction κ of the market value of assets, giving Fisherian debt deflation and a pecuniary externality.

Theory: a regulator committing to a policy would later have an incentive to break its promises; the authors solve for the time-consistent policy and prove that the macroprudential debt tax is positive.

Results:

- Crisis probability: $4\% \to 0.02\%$.
- Asset-price drop in a crisis: $-43.7\% \to -5.4\%$.
- Welfare: $+0.30\%$ of consumption.
- Simpler rules do much worse: best fixed tax 0.03%, best Taylor-type rule 0.09%.

Convincing on theory and solution; my comments are about the **calibration targets**.

Where the headline numbers come from

Key data object: a "crisis" is defined as a period in which the linearly detrended current account is more than 2 SD above its mean.

- 34 OECD countries, 1984–2012: frequency 4%, duration 1 year.

One number per parameter:

- 4% crisis frequency $\rightarrow$ how often credit tightens: $P_{H,L} = 0.1$.
- US mortgage loan-to-value ratios $\rightarrow$ how much it tightens: κ from 0.9 to 0.75.
- US data for 2013 $\rightarrow$ debt (β) and working capital (θ).

No inference: targets are treated as exact, and the model is solved and simulated.

Comments roadmap

Comment 1 on the 4% target; Comment 2 on secured vs unsecured credits; Comment 3 on the κ calibration.

Comment 1: The 4% target does not discipline crisis size

Concern: the 4% comes from a rule stated in standard deviations, and no target in Table 1 concerns the size of crises.

- Table 4 in the appendix lists the episodes considered as crises in 1984–2012. Notorious European crises connected to the Great Recession are notably absent (Greece, Italy, Ireland, Iceland, Portugal).
- The model's crises are very large: consumption -26%, asset prices -43.7%, credit down almost 18 pp of GDP.

Why it matters: the welfare gain comes from crises being rarer and milder, so it depends on whether the 4% are events of this size.

- The authors recognize that the model's reversal and consumption fall exceed US data in 2008.

Suggested fixes:

1. Redefine a crisis so that it can include notorious episodes (e.g. the Great Recession in the EU).
2. Add a crisis-size target for calibration.
3. Alternatively, select episodes by severity, including only the worst crises in the count.

Comment 2: All credits are treated as secured

Concern: the cap is calibrated on secured loans and treats all credit as secured by asset values, but only 40% of short-term commercial loans use collateral, and θ includes trade credit.

$$-\frac{b_{t+1}}{R_t} + \theta p_v v_t \leq \kappa_t q_t k_t \tag{3}$$

Why it matters: the authors explain Jeanne and Korinek's (2010) weak policy effect by pointing to the low share of borrowing tied to asset values, 7% there.

Suggested fixes:

1. Add a constant to the credit constraint, along the lines of Jeanne and Korinek, to reflect that a part of borrowing capacity does not move with asset prices.
2. Report results under the new model calibration.

Comment 3: κ^L is set at the low end

Concern: κ^L takes the bottom of the 0.75–0.8 range for US mortgage LTVs around 2008 and applies that drop to every crisis.

- $0.9 \to 0.75$ cuts borrowing capacity by 17% for fixed asset prices; $0.9 \to 0.8$ cuts it by 11%, a third less.

Why it matters: κ^L is the direct lever on crisis size, but it is never varied.

- Crises coincide with switches to κ^L.
- Results with κ variations are never reported.

Suggested fixes:

1. Report results for $\kappa^L = 0.8$, the top estimate cited, up from the minimum estimate of 0.75.
2. Alternatively, free κ^L to match a crisis-size target (Comment 1).

Extension: Do LTV tightenings do what the model says?

Idea: the paper treats LTV rules as similar to its debt tax, and in the model the optimal tax rises with debt. Do LTV tightenings restrain credit more when leverage is high?

What it would need:

- Dates of LTV rules (e.g. Finland, Hong Kong, Indonesia, New Zealand, Poland and South Korea are named by the paper).
- Credit and asset-price series, and crisis dates.

Method: panel local projections of credit, asset prices and crisis incidence on LTV tightenings, interacted with leverage.

Main difficulty: these rules aim to make credit costlier when it is expanding rapidly, so they need an instrument or narrative dates.

Minor comments

1. Fig. 7 stops at 2%, but the Conclusion ranks a 3.6% fixed tax below no policy.
2. Table 4 and Fig. F.3 disagree for Spain, Sweden, Switzerland, Turkey, the UK and the US.

Conclusion and questions for the room

Assessment: very nice paper: clean theory of time-consistent macroprudential policy, a model solved with a global nonlinear method, and the same crisis rule in the data and the unregulated economy.

Main concern: the 4% target does not discipline **crisis size**, all credits are treated as **secured**, and κ is set at the low bound.

Questions:

1. Would you let a regulator run a debt tax that is active in almost 94% of states on this evidence?
2. Are real regulators closer to commitment or to discretion?
3. What should count as a crisis when calibrating a model like this?